

Here $\mathcal{C}(j)$ is the set of children of bus j , and the asterisk superscript in (1) denotes complex conjugation. Actual line loss is $P_{\text{loss},t} = S_b \sum_{(i,j) \in E} r_{ij} |I_{ij,t}|^2$ MW. Inverter rating S_i^{inv} (MVA) determines its instantaneous reactive capability $q_{i,t}^{\text{max}}$ (Mvar). First clip the previous action to the present capability interval, defining $q_{i,t}^{\text{base}} = \text{clip}(q_{i,t-1}, -q_{i,t}^{\text{max}}, q_{i,t}^{\text{max}})$. The nominal ramp rate R_i (Mvar/h) and one-hour interval Δt constrain actions relative to this base. AC evaluation checks

$$q_{i,t}^{\text{max}} = \sqrt{(S_i^{\text{inv}})^2 - (P_{i,t}^{\text{PV}})^2}, |q_{i,t}| \leq q_{i,t}^{\text{max}}, |q_{i,t} - q_{i,t}^{\text{base}}| \leq R_i \Delta t, 0.95 \leq |V_{i,t}| \leq 1.05, |I_{ij,t}| \leq I_{ij}^{\text{max}}. \quad (2)$$

θ contains all trainable parameters. At inference, $\hat{q}_{i,t} = \hat{a}_{i,t} q_{i,t}^{\text{max}}$ converts network output to Mvar. The previous action is first clipped to the present capability interval, yielding the ramp reference $q_{i,t}^{\text{base}}$. The network proposal $\hat{q}_t$ is then projected onto the device intersection $\mathcal{Q}$. With projection Π and $\Lambda = \{1, 0.75, 0.5, 0.25, 0\}$, the safety-screen candidates are